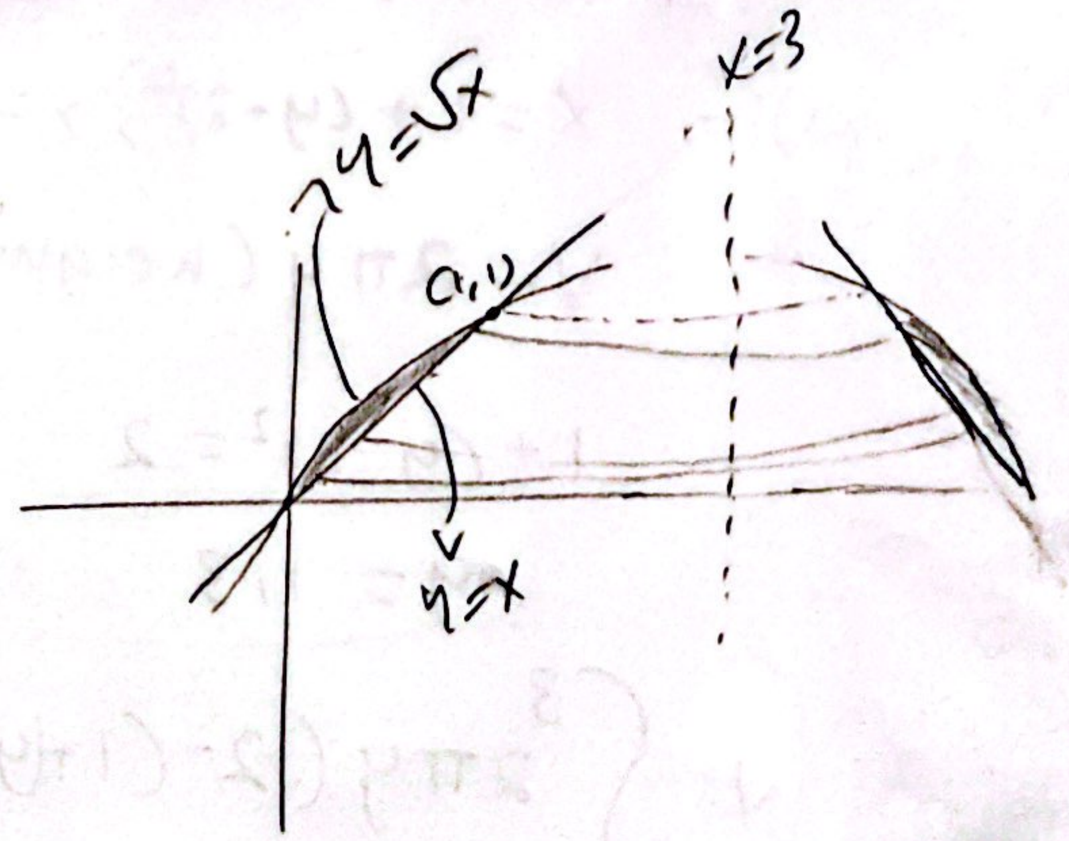
# Review #2 Solutions

1) Find the volume of the solid obtained by rotating the region bounded by the curves about $x=3$

$$y=x \ , \ y=\sqrt{x}$$




$x=\sqrt{x} \Rightarrow x=0,1$

Inner radius $= 3-y$

outer radius $= 3-y^2$

Cross-sectional area $= \pi(R^2-r^2)$

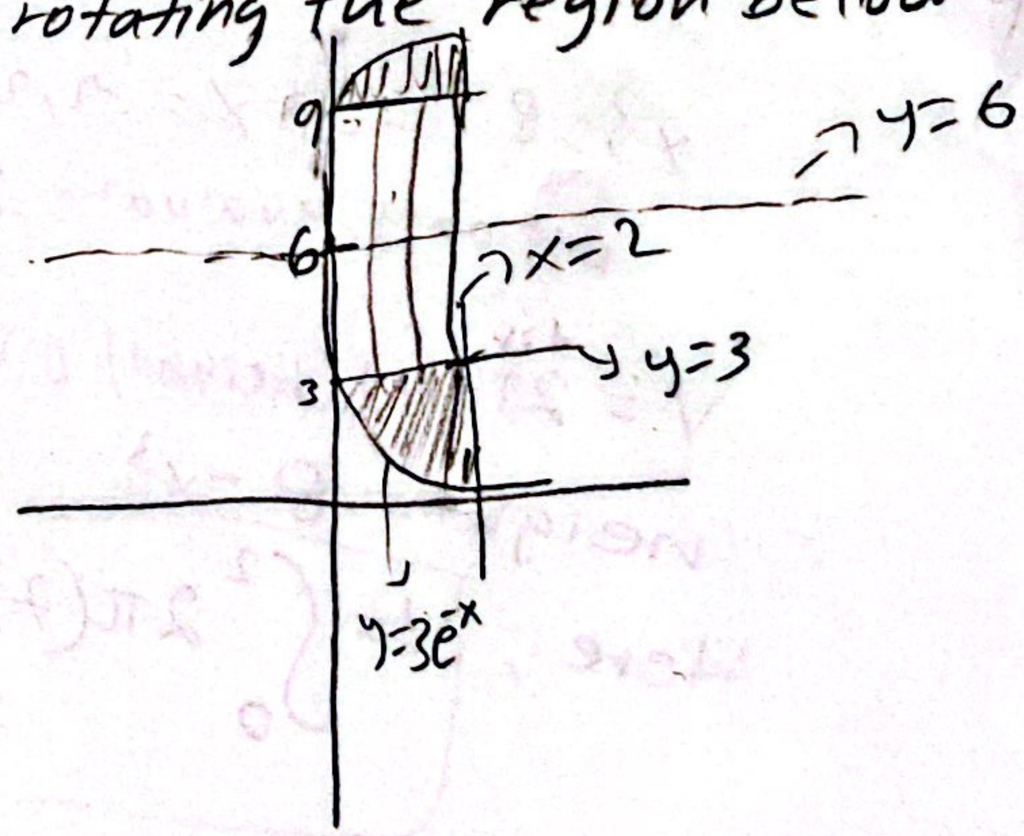
$$A=\pi((3-y^2)^2-(3-y)^2)$$

$$V=\int_0^1 \pi((3-y)^2-(3-y)^2\,dy$$

$$=\pi\left(\frac{y^5}{5}-\frac{7y^3}{3}+3y^2\right)\Big|_0^1=\boxed{\frac{13\pi}{15}}$$

2) Find the volume obtained by rotating the region below about $y=6$.

$$y=3e^{-x},\ y=3,\ x=2$$




$3e^{-x}=3 \Rightarrow x=0$

Inner radius $= 6-3=3$

outer radius $= 6-3e^{-x}$

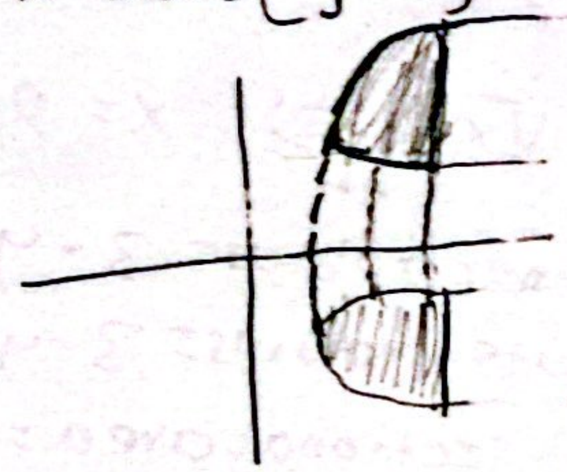
$$A=\pi((6-3e^{-x})^2-(6-3)^2)$$

$$\int_0^2 \pi((6-3e^{-x})^2-(3)^2)\,dx=\boxed{\frac{9\pi(5e^4+8e^2-1)}{2e^4}}$$

Scanned with CamScanner

3) Use methods of cylindrical shells to find the volume generated by rotating the following curves along the specified axes. (no calculation necessary)

a) $x = 1 + (y-2)^2$, $x = 2$ along x-axis [$y=0$]



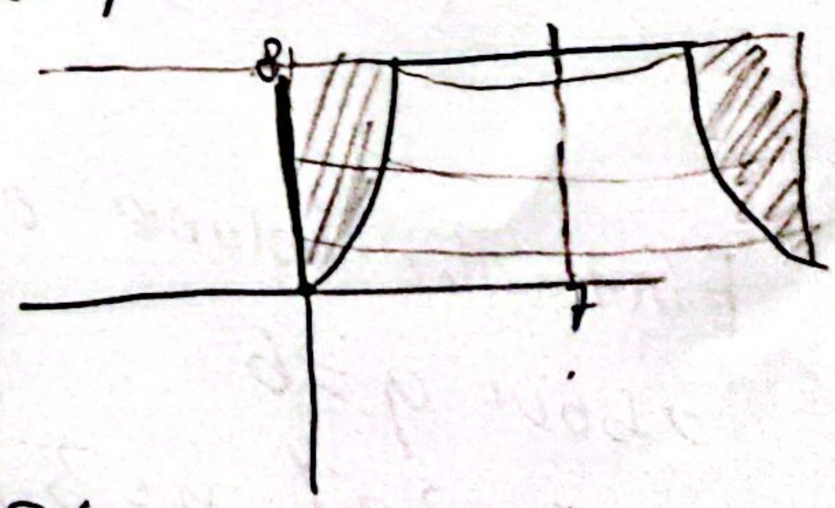
$V = 2\pi y\,(\text{height})(\Delta y)$

$1 + (y-2)^2 = 2$

$y = 1, 3$

$$\boxed{V = \int_1^3 2\pi y\left(2 - (1+(y-2)^2)\right) dy}$$

b) $y = x^3$, $y = 8$, $x = 0$ along $x = 7$




$x^3 = 8 \Rightarrow x = -2, 2$

we only evaluate from 0 to 2

$V = 2\pi(x)(\text{height})\,\Delta x$ for $x = 0$ (y-axis rotation)

height $= 8 - x^3$

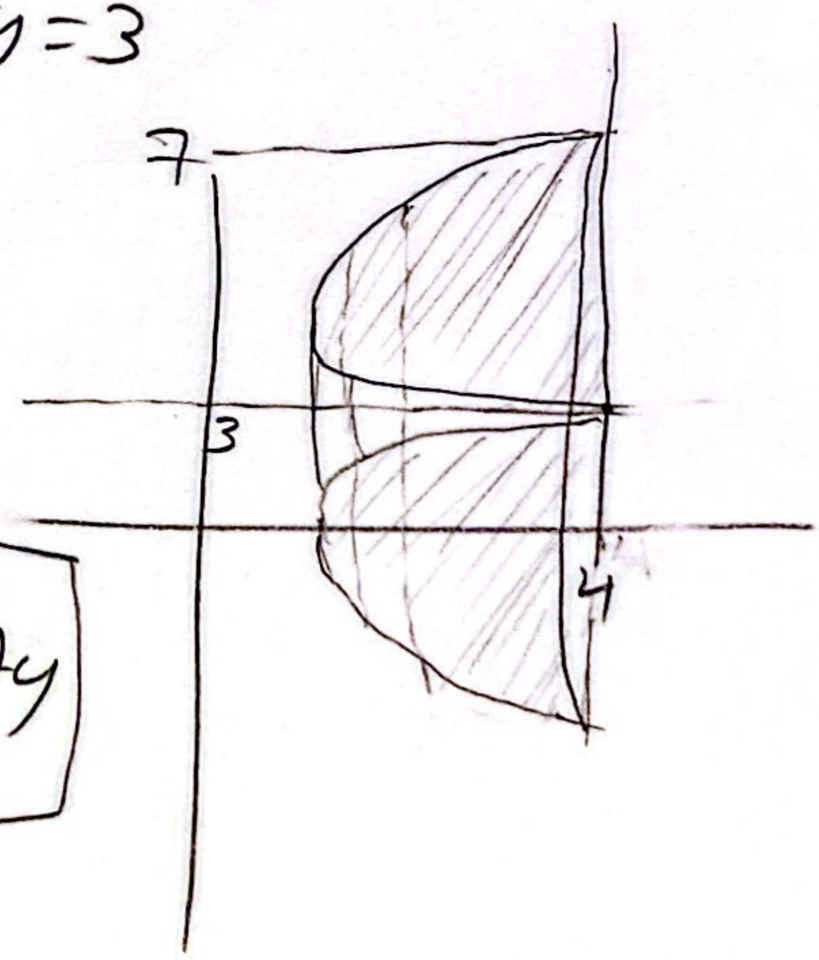
Here, $\boxed{V = \int_0^2 2\pi(7-x)(8-x^3)\,dx} = \frac{744\pi}{5}$

c) $(y-5)^2$, $x=4$ about $y=3$

$V = 2\pi(y)(\text{height})\Delta y$ for $y=0$

$V = 2\pi(y-3)(4-(y-5)^2)$

$(y-5)^2 = 4 \Rightarrow y = 3, 7$




Hence, $\boxed{V = \int_3^7 2\pi(y-3)(4-(y-5)^2)\,dy}$

$= \frac{128\pi}{3}$

Scanned with CamScanner